

The 50 Most Difficult Injuries to Detect — Volume 2

A Second Clinical Research Brief — New High-Miss-Risk Presentations (No Overlap with Volume 1)

Prepared for FisioterapiA clinical reasoning validation · 31 July 2026

Abstract

This second volume catalogues fifty additional high-miss-risk presentations that commonly defeat “obvious local injury” reasoning. Emphasis shifts toward pediatric hip emergencies (SCFE, septic arthritis), occult pelvic and carpal bone injury, dynamic ulnar wrist and elbow instability, nerve entrapments with normal sensation or normal canal MRI, vascular syndromes in athletes (effort thrombosis, popliteal entrapment), post-ACLR cyclops lesions, inflammatory girdle disease, dermatomal zoster without rash, and double-crush neuropathy. Together with Volume 1, the set supports a 100-case expert detection library for clinicians and clinical AI validation.

Method & scope

Entries were chosen to avoid duplication with Volume 1 titles while preserving the same structure: true pathology, cognitive trap, usual mislabel, ordered detection steps, and imaging/red-flag notes. Educational use only—not a substitute for emergency protocols or specialty guidelines.

Cross-cutting detection principles (Volume 2 additions)

- In adolescents with knee pain, examine the hip before committing to a knee pathway (SCFE).
- In radial head fractures, always clear the wrist/DRUJ (Essex-Lopresti).
- Dynamic symptoms (snapping ECU, PLRI apprehension) need dynamic exam/ultrasound—not only static MRI.
- Exertional limb pain in the young: think vascular entrapment and effort thrombosis, not only compartments/tendons.
- When canal MRI is “normal” but root pain is severe, demand far-lateral / exit-zone review.

Contents

Volume 2 expands into pediatric slips/infections, occult pelvic and carpal fractures, dynamic tendon instability, vascular entrapments, post-surgical cyclops lesions, inflammatory girdle disease, and multi-site “double crush” neuropathy. None of these fifty titles repeat Volume 1.

1. Slipped Capital Femoral Epiphysis (SCFE)
2. Septic Arthritis of the Hip (Child or Adult)
3. Sacral Insufficiency Fracture
4. Pubic Ramus / Pelvic Stress Fracture
5. Transient Osteoporosis of the Hip / Bone Marrow Edema Syndrome
6. Osteoid Osteoma
7. Soft-Tissue Sarcoma Misdiagnosed as Hematoma or “Muscle Tear”
8. Morel-Lavallée Lesion
9. Hook of Hamate Fracture
10. Stener Lesion (Complete UCL Thumb with Adductor Interposition)
11. Essex-Lopresti Injury
12. Posterolateral Rotatory Instability of the Elbow (PLRI)
13. Anterior Interosseous Nerve (AIN) Syndrome
14. Suprascapular Nerve Entrapment (Spinoglenoid / Suprascapular Notch)
15. Quadrilateral Space Syndrome
16. Long Thoracic Nerve Palsy (Serratus Anterior)
17. Spinal Accessory Nerve Injury
18. Effort Thrombosis (Paget–Schroetter Syndrome)
19. Popliteal Artery Entrapment Syndrome
20. Occult Tibial Plateau Fracture
21. Meniscal Ramp Lesion
22. Posterolateral Meniscal Root / Posterior Horn Subtle Tear Patterns
23. Cyclops Lesion (After ACL Reconstruction)
24. Isolated Subscapularis Tear
25. PASTA Lesion (Partial Articular Supraspinatus Tendon Avulsion)
26. Posterior Sternoclavicular Joint Dislocation
27. First Rib Stress Fracture
28. Slipping Rib Syndrome / Rib-Tip Syndrome
29. Maigne Syndrome (Thoracolumbar Junction Referral)
30. Bertolotti Syndrome (Lumbosacral Transitional Vertebra)
31. Extraforaminal Far-Lateral Disc Herniation (e.g., L5–S1 far-out)
32. Thoracic Disc Herniation
33. Cervical Artery Dissection Presenting as Neck Pain / Headache
34. Pulmonary Embolism Mimicking Musculoskeletal Chest/Scapular Pain
35. Herpes Zoster Sine Herpete (Radicular Pain Before or Without Rash)
36. Polymyalgia Rheumatica (PMR)
37. Calcaneal Stress Fracture
38. Plantar Plate Tear (Metatarsophalangeal)
39. Cuboid Syndrome
40. Os Trigonum / Posterior Ankle Impingement
41. ECU Subluxation / Instability
42. Intersection Syndrome
43. Wartenberg Syndrome (Superficial Radial Nerve)
44. Meralgia Paresthetica (Lateral Femoral Cutaneous Nerve)
45. Obturator Nerve Entrapment
46. Athletic Pubalgia / Inguinal Disruption (Sports Hernia Complex)
47. Osteitis Pubis
48. Baxter’s Nerve Entrapment (Inferior Calcaneal Nerve)
49. Complex Regional Pain Syndrome (CRPS) — Early
50. Double Crush Syndrome (Proximal + Distal Nerve Compromise)

The fifty injuries (Volume 2)

1. Slipped Capital Femoral Epiphysis (SCFE)

Pediatric / adolescent hip

What it actually is

Displacement of the femoral head epiphysis through the growth plate, typically in overweight adolescents.

Why it is tricky

Often presents as knee or thigh pain only; hip exam is skipped and the slip progresses.

Commonly mistaken for

Growing pains, patellofemoral pain, groin strain

Correct detection steps

- Any adolescent with limp + knee/thigh pain must have a hip exam.
- Limited internal rotation and obligate external rotation in flexion are classic.
- Bilateral AP pelvis + frog-lateral radiographs urgently.
- Non-weight-bearing and immediate ortho referral once suspected—do not “mobilize.”

Red flags

- Inability to bear weight
- Bilateral symptoms
- Endocrine risk factors

Imaging / confirmation

AP pelvis and frog-lateral X-rays; MRI if high suspicion and X-ray equivocal.

2. Septic Arthritis of the Hip (Child or Adult)

Infection emergency

What it actually is

Bacterial infection of the hip joint that rapidly destroys cartilage if delayed.

Why it is tricky

Early cases may lack dramatic fever; in children it overlaps with transient synovitis.

Commonly mistaken for

Transient synovitis, appendicitis (referred), muscle strain

Correct detection steps

- Non-weight-bearing, fever, irritability, or toxic appearance != infection pathway.
- Kocher/modified criteria in children (fever, non-weight-bearing, ESR/CRP, WBC).
- Urgent labs + ultrasound/MRI and orthopedic emergency referral for aspiration.
- Never inject or aggressively mobilize a hot irritable hip before infection is excluded.

Red flags

- Fever
 - Refusal to walk
 - Immunosuppression
-

3. Sacral Insufficiency Fracture

Occult bone injury

What it actually is

Fracture of the sacrum in osteoporotic or post-radiation bone, often without major trauma.

Why it is tricky

Presents as “severe lumbosacral pain”; plain lumbar X-rays miss it; labeled mechanical LBP.

Commonly mistaken for

Facet syndrome, SI mechanical pain, lumbar stenosis flare

Correct detection steps

- Older adult, osteoporosis, sudden buttock/sacral pain, difficulty walking.
- Focal sacral tenderness; pain with transfers more than with classic disc maneuvers.
- MRI or CT of the pelvis/sacrum—not only lumbar MRI.
- Protect loading; bone-health work-up and specialist input.

Imaging / confirmation

MRI preferred; CT if MRI unavailable; X-ray insensitive.

4. Pubic Ramus / Pelvic Stress Fracture

Occult bone injury

What it actually is

Stress fracture of the pubic rami or pelvic ring in runners, military recruits, or osteoporotic patients.

Why it is tricky

Groin pain is treated as adductor tendinopathy; athletes keep training.

Commonly mistaken for

Adductor strain, osteitis pubis, inguinal hernia

Correct detection steps

- Progressive groin/perineal pain with impact; one-leg stance or hop painful.
- Point bony tenderness on pubic ramus; adductor squeeze may hurt but bone pain dominates.
- MRI when suspicion high despite normal X-ray.
- Relative rest; rule out femoral neck stress fracture in the same athlete phenotype.

Imaging / confirmation

MRI; X-ray often normal early.

5. Transient Osteoporosis of the Hip / Bone Marrow Edema Syndrome

Occult bone / marrow

What it actually is

Acute bone marrow edema of the femoral head/neck causing severe hip pain, usually self-limited but easily confused with AVN.

Why it is tricky

Looks like AVN or “severe tendinopathy”; wrong labeling changes urgency and load advice.

Commonly mistaken for

AVN, stress fracture, synovitis

Correct detection steps

- Acute/subacute deep groin pain, marked antalgic gait, often middle-aged men or pregnancy-associated in women.
- Severe pain with IR and weight-bearing; X-ray may be normal early or show osteopenia later.
- MRI distinguishes diffuse marrow edema pattern from classic AVN demarcation—radiology liaison matters.
- Protect weight-bearing until imaging clarifies; do not assume “soft tissue only.”

Imaging / confirmation

MRI is essential for differentiation from AVN/stress fracture.

6. Osteoid Osteoma

Tumor — often missed

What it actually is

Benign bone-forming tumor with intense night pain typically relieved by NSAIDs.

Why it is tricky

Pain is blamed on “growing pains,” tendinopathy, or psychogenic causes for months–years.

Commonly mistaken for

Night-pain tendinopathy, osteomyelitis, stress fracture

Correct detection steps

- Focal bone pain worse at night, dramatic NSAID relief is a classic clue.
- Normal early X-ray does not exclude—CT often shows nidus best.
- Do not endless-treat as soft tissue if night bone pain persists.
- Refer orthopedics/oncology pathways for confirmation and ablation/excision options.

Imaging / confirmation

CT for nidus; MRI/bone scan as adjuncts.

7. Soft-Tissue Sarcoma Misdiagnosed as Hematoma or “Muscle Tear”

Oncology red flag

What it actually is

Malignant soft-tissue tumor presenting as a lump or “strain that never heals.”

Why it is tricky

History of minor trauma invites hematoma labeling; mass grows quietly.

Commonly mistaken for

Hematoma, muscle strain, lipoma, abscess

Correct detection steps

- Any deep mass >5 cm, growing, firm, or painless deep swelling needs tumor pathway—not repeated aspiration.
- Night pain, systemic symptoms, or recurrence after “hematoma” treatment raise concern.
- Urgent ultrasound/MRI via sarcoma referral guidelines; avoid unplanned excision.

Red flags

- Growing mass
 - Deep firm lump
 - Constitutional symptoms
-

8. Morel-Lavallée Lesion

Closed degloving soft-tissue

What it actually is

Closed shear degloving injury creating a hemolymphatic collection between subcutaneous fat and fascia (often hip/thigh/knee).

Why it is tricky

Looks like a simple bruise or bursitis; fluctuance and delayed swelling are missed.

Commonly mistaken for

Contusion, trochanteric bursitis, hematoma

Correct detection steps

- Shearing trauma (motorcycle, sports) + fluctuant fluid collection over greater trochanter/thigh.
- Ultrasound/MRI confirms encapsulated collection.
- Refer for appropriate drainage/compression—do not treat as ordinary bruise indefinitely.

Imaging / confirmation

Ultrasound or MRI.

9. Hook of Hamate Fracture

Occult wrist fracture

What it actually is

Fracture of the hamate hook, common in bat/racket/golf sports from direct impact or repetitive stress.

Why it is tricky

Standard wrist X-rays miss it; labeled ulnar-sided wrist sprain or ECU tendinopathy.

Commonly mistaken for

ECU tendinopathy, TFCC strain, pisotriquetral arthritis

Correct detection steps

- Ulnar palm pain in grip sports; pain over Guyon's/hamate region.
- Pain with resisted flexion of 4th/5th digits; grip weakness.
- Request carpal tunnel view or CT dedicated to hamate if clinical suspicion high.
- Early diagnosis prevents nonunion and ulnar nerve/flexor irritation.

Imaging / confirmation

Carpal tunnel X-ray view; CT is often definitive.

10. Stener Lesion (Complete UCL Thumb with Adductor Interposition)

Thumb UCL — surgical

What it actually is

Complete rupture of the thumb MCP ulnar collateral ligament where the adductor aponeurosis blocks healing.

Why it is tricky

All “skier’s thumb” injuries get the same splint advice; Stener lesions need surgery.

Commonly mistaken for

Partial UCL sprain, MCP capsular sprain

Correct detection steps

- Valgus injury to thumb; tenderness at UCL; laxity without firm end-feel.
- Palpable displaced stump sometimes present; compare to contralateral side carefully.
- Ultrasound/MRI to assess displacement/Stener anatomy.
- Urgent hand surgery referral for complete/Stener—do not prolonged “buddy tape only.”

Imaging / confirmation

Ultrasound or MRI; stress X-rays selectively by specialist.

11. Essex-Lopresti Injury

Forearm longitudinal instability

What it actually is

Radial head fracture + interosseous membrane disruption + DRUJ injury causing longitudinal forearm instability.

Why it is tricky

Attention stays on the radial head; wrist/DRUJ pain is dismissed until the radius migrates proximally.

Commonly mistaken for

Isolated radial head fracture, wrist sprain

Correct detection steps

- Always examine the wrist/DRUJ in every radial head fracture.
- Wrist pain, DRUJ instability, or radiographic ulnar-positive variance change after elbow injury.
- Compare bilateral wrist films; consider MRI/CT for IOM.
- Early specialist management—delayed recognition is hard to salvage.

Imaging / confirmation

Elbow + wrist X-rays; MRI for interosseous membrane.

12. Posterolateral Rotatory Instability of the Elbow (PLRI)

Elbow instability — subtle

What it actually is

Insufficiency of the lateral ulnar collateral ligament complex allowing rotatory subluxation of the ulnohumeral joint.

Why it is tricky

Patients report clicking/apprehension with push-up or chair rise; standard “tennis elbow” care fails.

Commonly mistaken for

Lateral epicondylitis, radial tunnel, loose body

Correct detection steps

- History of elbow dislocation/sprain or corticosteroid injection near LUCL.
 - Lateral pivot-shift / tabletop / chair push-up apprehension.
 - MRI for LUCL; exam under anesthesia sometimes needed.
 - Stop repeated epicondylar injections if instability signs exist.
-

13. Anterior Interosseous Nerve (AIN) Syndrome

Peripheral nerve — pure motor

What it actually is

Motor neuropathy of the AIN causing weakness of FPL, FDP to index (and often PQ) with no sensory loss.

Why it is tricky

Inability to make an “OK” sign is blamed on tendon rupture of FPL.

Commonly mistaken for

FPL tendon rupture, brachial neuritis (can overlap), C8 root

Correct detection steps

- Pure motor deficit: cannot flex thumb IP and index DIP; sensation intact.
 - Tenodesis effect intact if tendon continuous—distinguishes from rupture.
 - Consider Parsonage–Turner spectrum when pain precedes weakness.
 - EMG localization; hand/neuro referral.
-

14. Suprascapular Nerve Entrapment (Spinoglenoid / Suprascapular Notch)

Peripheral nerve at shoulder

What it actually is

Compression of the suprascapular nerve (notch !' supra- and infraspinatus; spinoglenoid !' infraspinatus mainly), often by cyst from labral tear.

Why it is tricky

Infraspinatus atrophy and weak external rotation are labeled “cuff tear” indefinitely.

Commonly mistaken for

Infraspinatus tear, C5–C6 radiculopathy, Parsonage–Turner

Correct detection steps

- Vague posterolateral shoulder pain + isolated infra ± supra atrophy.
- Weak ER with relatively quiet subacromial tests.
- MRI for paralabral cyst / notch anatomy; EMG confirms.
- Treat labral/cyst source—not endless cuff strengthening alone.

Imaging / confirmation

Shoulder MRI; EMG.

15. Quadrilateral Space Syndrome

Axillary nerve entrapment

What it actually is

Compression of the axillary nerve (and posterior humeral circumflex artery) in the quadrilateral space.

Why it is tricky

Poorly localized shoulder pain in throwers; cuff imaging often “almost normal.”

Commonly mistaken for

Posterior cuff tendinopathy, thoracic outlet, referred cervical pain

Correct detection steps

- Thrower/overhead athlete with poorly localized posterior shoulder pain and fatigue.
- Point tenderness over quadrilateral space; delayed deltoid/teres minor weakness or paresthesia.
- MRI may show teres minor denervation edema; angiography selectively for vascular form.
- Specialist sports shoulder pathway if persistent.

16. Long Thoracic Nerve Palsy (Serratus Anterior)

Peripheral nerve

What it actually is

Injury/neuritis of the long thoracic nerve causing serratus anterior weakness and medial scapular winging.

Why it is tricky

Labeled “scapular dyskinesis” or poor posture; true winging and nerve cause are missed.

Commonly mistaken for

Scapular dyskinesis, trapezius palsy, cuff weakness

Correct detection steps

- Medial border winging worse with wall push-up / forward flexion.
- Distinguish from spinal accessory (trapezius) winging pattern.
- EMG of serratus; MRI cervical/brachial plexus if needed to exclude structural causes.
- Protect from fatigue; neurology/ortho referral for persistent deficits.

17. Spinal Accessory Nerve Injury

Peripheral nerve — iatrogenic risk

What it actually is

Injury to CN XI causing trapezius weakness (often after lymph node biopsy or trauma).

Why it is tricky

Shoulder droop and pain are treated as cuff disease; surgical scar history is ignored.

Commonly mistaken for

Rotator cuff tear, long thoracic palsy, adhesive capsulitis

Correct detection steps

- Ask about neck surgery/biopsy trauma.
 - Shoulder droop, asymmetric neckline, weak shrug, lateral winging pattern.
 - EMG of trapezius; early recognition improves reconstructive options.
-

18. Effort Thrombosis (Paget–Schroetter Syndrome)

Vascular emergency

What it actually is

Primary upper-extremity deep vein thrombosis of the subclavian vein, often in young overhead athletes.

Why it is tricky

Arm “pump” swelling after training is called muscle hypertrophy or TOS soft-tissue pain.

Commonly mistaken for

TOS (neurogenic), biceps strain, allergic angioedema

Correct detection steps

- Acute/subacute unilateral arm swelling, heaviness, cyanosis after repetitive overhead effort.
- Visible venous collaterals over shoulder/chest.
- Urgent duplex ultrasound and vascular referral—this is not “rest and ice.”

Red flags

- Rapid arm swelling
- Discoloration
- Chest pain/dyspnea (PE)

19. Popliteal Artery Entrapment Syndrome

Vascular — exertional

What it actually is

Anatomic or functional compression of the popliteal artery by gastrocnemius/medial head variants causing exertional calf ischemia.

Why it is tricky

Young athletes with calf pain are labeled CECS or shin splints; pulses at rest may be normal.

Commonly mistaken for

CECS, tibial stress fracture, medial gastroc strain

Correct detection steps

- Exertional calf pain/paresthesia in young patient; symptoms with resisted plantarflexion/active ankle maneuvers.
- Pulse diminish with provocative ankle positions.
- Duplex with provocation / MRA / CTA via vascular specialist.
- Do not proceed to compartment release without vascular exclusion when clues fit.

Imaging / confirmation

Provocation duplex, MRA/CTA.

20. Occult Tibial Plateau Fracture

Occult bone injury

What it actually is

Minimally displaced tibial plateau fracture after valgus/axial load; may be invisible on initial films.

Why it is tricky

Knee “sprain” with inability to bear weight; ligament testing is done on an unstable fracture.

Commonly mistaken for

MCL sprain, meniscal tear, contusion

Correct detection steps

- Age + axial/valgus trauma + effusion + inability to walk !' image before aggressive stress testing.
- Lipohemarthrosis on X-ray/CT is a clue to intra-articular fracture.
- CT if X-ray negative but high suspicion; MRI for occult fracture and soft tissue.

Imaging / confirmation

X-ray then CT/MRI as needed.

21. Meniscal Ramp Lesion

Knee — frequently missed on MRI

What it actually is

Meniscocapsular tear of the posterior horn medial meniscus, strongly associated with ACL rupture.

Why it is tricky

Standard MRI reports may miss it; residual instability persists after ACL reconstruction if untreated.

Commonly mistaken for

Simple medial meniscus tear, posteromedial pain from MCL

Correct detection steps

- ACL injury + posteromedial joint-line pain / instability sensations.
- Ask radiology specifically about ramp; arthroscopic probing remains gold standard when suspected.
- Ortho ACL pathway should routinely assess ramp lesions.

Imaging / confirmation

High-quality MRI; intraoperative confirmation.

22. Posterolateral Meniscal Root / Posterior Horn Subtle Tear Patterns

Knee cartilage–meniscus

What it actually is

Subtle posterior horn/root pathology of the lateral meniscus, sometimes with ACL injury, easy to under-call.

Why it is tricky

Mechanical symptoms are intermittent; MRI subtlety leads to “normal knee” labels.

Commonly mistaken for

IT band syndrome, patellofemoral pain, plica

Correct detection steps

- Deep joint-line pain, squatting pain, subtle locking after pivot injury.
 - Correlate MRI with clinical mechanical signs; seek second radiology read if mismatch.
 - Ortho review when effusion + mechanical symptoms persist.
-

23. Cyclops Lesion (After ACL Reconstruction)

Post-surgical complication

What it actually is

Nodular fibroproliferative tissue anterior in the notch causing extension block after ACLR.

Why it is tricky

Blamed on “quad inhibition” or arthrofibrosis generally; specific lesion is treatable arthroscopically.

Commonly mistaken for

Extension deficit from pain inhibition, graft impingement, arthrofibrosis diffuse

Correct detection steps

- Loss of terminal extension weeks–months after ACLR, anterior clunk.
- MRI shows anterior notch nodule.
- Early ortho arthroscopy discussion if extension does not progress with rehab.

Imaging / confirmation

MRI knee.

24. Isolated Subscapularis Tear

Shoulder — under-detected cuff tear

What it actually is

Tear of subscapularis, often with biceps subluxation, missed on casual cuff screening focused on supra/infra.

Why it is tricky

Patients can still elevate the arm; lift-off/belly-press are skipped.

Commonly mistaken for

Biceps tendinopathy, anterior cuff tendinopathy, capsulitis

Correct detection steps

- Anterior shoulder pain, weak internal rotation, positive bear-hug/belly-press/lift-off as able.
- Look for biceps instability signs.
- MRI/ultrasound with specific subscapularis attention.
- Early repair consideration for full-thickness tears—timing matters.

Imaging / confirmation

MRI or high-quality ultrasound.

25. PASTA Lesion (Partial Articular Supraspinatus Tendon Avulsion)

Shoulder — partial cuff

What it actually is

Partial-thickness articular-sided supraspinatus tear that behaves like refractory “impingement.”

Why it is tricky

Bursal-sided exam and subacromial injections give partial relief; articular lesion persists.

Commonly mistaken for

Subacromial pain syndrome, bursitis alone

Correct detection steps

- Overhead athlete/worker with persistent pain despite bursal care.
- MRI/MRA for articular partial tear grading.
- Escalate imaging when 3–6 months of quality rehab fails and strength/pain remain discrepant.

Imaging / confirmation

MRI; MR arthrogram improves articular-sided detection.

26. Posterior Sternoclavicular Joint Dislocation

Trauma emergency

What it actually is

Posterior displacement of the medial clavicle threatening trachea, vessels, and esophagus.

Why it is tricky

Looks like a medial clavicle sprain; vital structure compression can be catastrophic.

Commonly mistaken for

SC sprain, medial clavicle physeal injury in teens, costochondritis

Correct detection steps

- Trauma to shoulder/chest + medial clavicle pain; dysphagia, stridor, venous congestion are emergencies.
- Do not attempt blind reduction in clinic.
- CT angiography pathway and emergency specialist care.

Red flags

- Dyspnea
 - Stridor
 - Swallowing difficulty
 - Arm swelling/vascular signs
-

27. First Rib Stress Fracture

Occult bone injury — overhead athletes

What it actually is

Stress fracture of the first rib in throwers, rowers, or laborers with repetitive scapular load.

Why it is tricky

Vague shoulder/neck/upper chest pain; standard shoulder MRI may omit the first rib.

Commonly mistaken for

TOS, scalene strain, cuff tendinopathy

Correct detection steps

- Insidious upper thoracic/shoulder girdle pain in overhead athletes.
- Pain with deep breath, ipsilateral side-bend, or scapular loading.
- Dedicated rib radiographs/CT/bone scan/MRI as indicated.
- Relative rest from causative load.

Imaging / confirmation

CT or MRI/bone scan; plain films may miss early lesions.

28. Slipping Rib Syndrome / Rib-Tip Syndrome

Chest wall — underdiagnosed

What it actually is

Hypermobility of false ribs (often 8–10) irritating intercostal nerves, causing sharp flank/upper abdominal pain.

Why it is tricky

Triggers GI work-ups for years; MSK cause is never examined.

Commonly mistaken for

Cholecystitis, splenic flexure syndrome, intercostal strain

Correct detection steps

- Sharp positional pain under rib margin, “popping” sensation, reproducible with hooking maneuver by trained clinician.
- Pain reproducible to palpation of rib tip; GI red flags still screened.
- Dynamic ultrasound sometimes helpful; specialist chest-wall/MSK referral.

29. Maigne Syndrome (Thoracolumbar Junction Referral)

Referred spinal pain

What it actually is

Dysfunction/irritation at T12–L2 referring to low back, iliac crest, groin, or trochanter via posterior rami / cluneal nerves.

Why it is tricky

Patient and clinician focus on the lumbosacral pain site; the thoracolumbar junction is never examined.

Commonly mistaken for

Facet L4–L5 pain, SI pain, GTPS, inguinal strain

Correct detection steps

- Iliac crest/buttock pain with tenderness of the iliac crest skin rolling (posterior rami territory).
 - Examine thoracolumbar junction motion and provocation—not only L5–S1.
 - If local LS treatments fail repeatedly, reassess T12–L2.
-

30. Bertolotti Syndrome (Lumbosacral Transitional Vertebra)

Congenital variant with pain generator

What it actually is

Pain associated with a lumbosacral transitional vertebra and anomalous articulation (often L5 transverse process–sacrum).

Why it is tricky

Variant is called “incidental”; true Bertolotti pain is under-recognized and over-called equally—needs careful linkage.

Commonly mistaken for

Facet syndrome, SI pain, discogenic LBP

Correct detection steps

- Younger patient with unilateral lumbosacral pain; imaging shows transitional anatomy.
- Correlate side of anomaly with pain; diagnostic injection sometimes used by specialists.
- Do not attribute all LBP to Bertolotti—but do not ignore a matching pattern either.

Imaging / confirmation

Lumbar radiographs/CT/MRI showing transitional vertebra.

31. Extraforaminal Far-Lateral Disc Herniation (e.g., L5–S1 far-out)

Radiculopathy — MRI can miss if not looked for

What it actually is

Disc herniation lateral to the foramen compressing the exiting root (e.g., L5 at L5–S1 far-lateral).

Why it is tricky

Standard MRI focus on the canal looks “normal”; severe root pain persists.

Commonly mistaken for

Hip OA, GTPS, peroneal palsy, “MRI-negative sciatica”

Correct detection steps

- Severe root pain with negative canal reading—ask specifically about far-lateral/exit zones.
- L5 signs with L5–S1 far-lateral herniation pattern.
- Coronal/foraminal sequences or CT; specialist radiology review.

Imaging / confirmation

MRI with attention to extraforaminal zones.

32. Thoracic Disc Herniation

Central / thoracic neurologic

What it actually is

Thoracic disc prolapse causing band-like chest/abdominal pain or cord signs.

Why it is tricky

Band-like pain triggers cardiac/GI pathways; thoracic spine is rarely examined.

Commonly mistaken for

Intercostal neuralgia, cholecystitis, cardiac pain, shingles pre-rash

Correct detection steps

- Band-like trunk pain, worse with cough/sneeze; examine thoracic cord signs (gait, reflexes).
- If myelopathic signs !' urgent MRI thoracic spine.
- Still exclude cardiac/visceral emergencies first when appropriate.

Red flags

- Gait change
 - Hyperreflexia
 - Bowel/bladder change
-

33. Cervical Artery Dissection Presenting as Neck Pain / Headache

Vascular neurologic emergency

What it actually is

Dissection of carotid or vertebral artery presenting with neck pain, headache, or Horner's—before stroke is obvious.

Why it is tricky

Attributed to “cervical strain” after minor trauma or sudden movement; manipulation risk context.

Commonly mistaken for

Mechanical neck pain, migraine, musculoskeletal headache

Correct detection steps

- Sudden unusual neck pain/headache, especially unilateral, $\pm$ Horner's, pulsatile tinnitus, or neuro symptoms.
- Any focal neuro sign !' emergency pathway (CTA/MRA), not physio progression.
- Avoid end-range cervical thrust when dissection possible.

Red flags

- Horner's syndrome
- Acute focal neurology
- Thunderclap or unique severe headache

34. Pulmonary Embolism Mimicking Musculoskeletal Chest/Scapular Pain

Medical emergency

What it actually is

PE presenting with pleuritic chest, scapular, or upper-back pain without obvious dyspnea at rest.

Why it is tricky

Patients seek care for “rib strain” after travel or surgery.

Commonly mistaken for

Rib sprain, costochondritis, scapular myalgia

Correct detection steps

- Risk factors: recent surgery, immobility, cancer, prior VTE, OCP, leg swelling.
- Pleuritic pain, tachycardia, breathlessness on exertion, hemoptysis.
- Wells/PERC-informed urgent medical assessment—not massage of the scapula.

Red flags

- Syncope
- Hypoxia
- Hemoptysis
- Unilateral leg swelling

35. Herpes Zoster Sine Herpete (Radicular Pain Before or Without Rash)

Infectious neuropathic

What it actually is

Varicella-zoster reactivation causing severe dermatomal pain with delayed, minimal, or absent rash.

Why it is tricky

Treated as radiculopathy or “muscle spasm” until vesicles appear—or they never clearly do.

Commonly mistaken for

Cervical/lumbar radiculopathy, intercostal strain

Correct detection steps

- Acute unilateral burning dermatomal pain out of proportion, allodynia, older/immunocompromised patient.
- Inspect skin daily; early antiviral window if vesicles appear.

- Consider zoster in severe unexplained dermatomal pain.

36. Polymyalgia Rheumatica (PMR)

Inflammatory — shoulder/hip girdle

What it actually is

Inflammatory condition of older adults with bilateral shoulder/hip girdle stiffness and pain, dramatically steroid-responsive.

Why it is tricky

Labeled bilateral cuff tendinopathy or “fibromyalgia onset.”

Commonly mistaken for

Bilateral rotator cuff disease, cervical spondylosis, hypothyroidism myalgia

Correct detection steps

- Age >50, bilateral shoulder ± pelvic girdle morning stiffness >45 minutes, systemic symptoms.
- Check ESR/CRP; screen for giant cell arteritis symptoms (headache, jaw claudication, vision).
- Rheumatology pathway—do not endless cuff rehab alone.

Red flags

- Jaw claudication
- Visual disturbance
- New temporal headache

37. Calcaneal Stress Fracture

Occult bone injury — hindfoot

What it actually is

Stress fracture of the calcaneus in runners, military recruits, or osteoporotic patients.

Why it is tricky

Heel pain is almost automatically plantar fasciitis.

Commonly mistaken for

Plantar fasciitis, fat-pad contusion, Sever's in teens

Correct detection steps

- Squeeze test of calcaneus positive; pain more posterior/body of calcaneus than fascia origin.
- Impact progression history; night pain possible.
- MRI if X-ray negative; unload early.

Imaging / confirmation

MRI; X-ray late sclerosis/line.

38. Plantar Plate Tear (Metatarsophalangeal)

Forefoot — underdiagnosed

What it actually is

Tear of the plantar plate at the MTP (often 2nd), causing pain, swelling, and progressive toe deformity.

Why it is tricky

Called metatarsalgia or Morton's neuroma; toe alignment slowly worsens.

Commonly mistaken for

Morton's neuroma, capsulitis, stress fracture of metatarsal

Correct detection steps

- Pain under MTP, swelling, positive drawer of toe, starting crossover toe deformity.
- Ultrasound/MRI for plantar plate integrity.
- Protect toe extension stress; foot & ankle specialist if unstable.

Imaging / confirmation

Ultrasound or MRI.

39. Cuboid Syndrome

Midfoot — clinical diagnosis

What it actually is

Painful disruption of cuboid position/mobility, often after inversion ankle injury or overuse.

Why it is tricky

X-rays normal; residual lateral midfoot pain after “resolved sprain.”

Commonly mistaken for

Peroneal tendinopathy, 5th MT stress reaction, sinus tarsi pain

Correct detection steps

- Lateral midfoot pain, pain with push-off, tenderness over cuboid.
- Often follows inversion sprain; peroneal tightness common.
- Clinical diagnosis; imaging excludes fracture. Trial of cuboid mobilization by skilled clinician.

40. Os Trigonum / Posterior Ankle Impingement

Posterior ankle — dancers/soccer

What it actually is

Painful posterior impingement from os trigonum or Stieda process, often with FHL irritation.

Why it is tricky

Labeled Achilles tendinopathy; pain is deeper and provoked by plantarflexion.

Commonly mistaken for

Achilles insertional tendinopathy, posterior ankle sprain, FHL tenosynovitis alone

Correct detection steps

- Plantarflexion pain in dancers/soccer players; posteromedial/lateral ankle pain.
- Pain with forced PF; FHL tunnel signs may coexist.
- Lateral X-ray/MRI for os trigonum and soft-tissue edema.

Imaging / confirmation

X-ray + MRI.

41. ECU Subluxation / Instability

Ulnar wrist — dynamic

What it actually is

Disruption of the ECU subsheath allowing the tendon to snap over the ulnar groove.

Why it is tricky

Static MRI may look nearly normal; patient describes painful snap with rotation.

Commonly mistaken for

TFCC tear, ulnar styloid fracture, ECU tendinopathy without instability

Correct detection steps

- Painful snap on ulnar wrist with supination/ulnar deviation (racquet sports).
- Palpate ECU during rotation—reproduce subluxation.
- Dynamic ultrasound is highly useful; MRI for associated TFCC.

Imaging / confirmation

Dynamic ultrasound ± MRI.

42. Intersection Syndrome

Forearm / wrist tendinopathy — location trap

What it actually is

Tenosynovitis where 1st and 2nd dorsal compartments cross (~4 cm proximal to Lister's).

Why it is tricky

Confused with De Quervain, which is more distal at the radial styloid.

Commonly mistaken for

De Quervain's tenosynovitis, radial sensory neuritis

Correct detection steps

- Rowers/weightlifters with pain/crepitus proximal to the wrist on the radial forearm.
- Tenderness at intersection site, not primarily at styloid.
- Ultrasound confirms; activity modification and load management.

43. Wartenberg Syndrome (Superficial Radial Nerve)

Peripheral nerve — sensory

What it actually is

Entrapment/neuritis of the superficial sensory branch of the radial nerve causing dorsal-radial hand dysesthesia.

Why it is tricky

Called De Quervain or "cervical"; tight watch/brace history is missed.

Commonly mistaken for

De Quervain, C6 radiculopathy, intersection syndrome

Correct detection steps

- Burning/numbness over dorsal first web without true motor loss.
- Tinel along radial sensory course; aggravation by ulnar flexion stretch.
- Remove compressive straps; differentiate from De Quervain Finkelstein-local pain.

44. Meralgia Paresthetica (Lateral Femoral Cutaneous Nerve)

Peripheral nerve — sensory

What it actually is

Compression of the LFCN under the inguinal ligament causing lateral thigh burning/numbness.

Why it is tricky

Labeled L2–L3 radiculopathy or "hip bursitis."

Commonly mistaken for

Lumbar radiculopathy, GTPS, hip OA referral

Correct detection steps

- Pure sensory lateral thigh symptoms; no motor deficit.
 - Worse with tight belts, pregnancy, weight change; pelvic compression may ease.
 - Normal hip motor exam; lumbar motor/reflex screen still done once.
 - Usually clinical diagnosis; imaging if atypical or progressive.
-

45. Obturator Nerve Entrapment

Peripheral nerve — medial thigh

What it actually is

Entrapment of the obturator nerve causing medial thigh pain and adductor weakness/paresthesia, seen in athletes after pelvic trauma or overuse.

Why it is tricky

Medial thigh pain is called chronic adductor strain; nerve signs are subtle.

Commonly mistaken for

Adductor tendinopathy, osteitis pubis, inguinal disruption

Correct detection steps

- Exercise-related medial thigh pain $\pm$ sensory change; adductor weakness out of proportion to tendon tenderness.
- Pain may be deep and poorly localized vs focal enthesis pain.
- MRI pelvis/EMG selectively; sports hip/groin specialist.

46. Athletic Pubalgia / Inguinal Disruption (Sports Hernia Complex)

Groin — overlapping entities

What it actually is

Spectrum of injuries to the inguinal canal posterior wall, conjoined tendon, and rectus/adductor aponeurotic plate without a true hernia bulge.

Why it is tricky

Normal hernia exam; athletes bounce between “adductor strain” and “hip impingement” labels.

Commonly mistaken for

Adductor strain, FAI/labral pain, true inguinal hernia, osteitis pubis

Correct detection steps

- Sports cutting/sprinting; pain with sit-up and resisted adductor together.
- Examine inguinal canal, pubic plate, hip ROM/FAI signs—map the pain generator(s).
- MRI protocols for athletic pubalgia; surgical opinion if recalcitrant.

Imaging / confirmation

Dedicated groin MRI.

47. Osteitis Pubis

Pubic symphysis overuse / inflammation

What it actually is

Inflammatory/overuse condition of the pubic symphysis and adjacent bone, common in kicking/cutting sports.

Why it is tricky

Overlaps with adductor enthesopathy and athletic pubalgia; imaging changes are nonspecific if clinical linkage is weak.

Commonly mistaken for

Adductor tear, infection of symphysis (rare but serious), stress fracture

Correct detection steps

- Central pubic pain, pain with squeeze test, kicking sports.
- Exclude infection if fever/systemic signs.
- MRI for bone-edema pattern; graded return-to-run/kick program.

Red flags

- Fever
- IVDU

- Severe constant night pain (infection/tumor)

48. Baxter's Nerve Entrapment (Inferior Calcaneal Nerve)

Heel pain — neuropathic

What it actually is

Entrapment of the first branch of the lateral plantar nerve causing medial heel pain, sometimes with abductor digiti minimi denervation.

Why it is tricky

Indistinguishable from plantar fasciitis at first glance; fails fascia-only care.

Commonly mistaken for

Plantar fasciitis, fat-pad syndrome, calcaneal stress fracture

Correct detection steps

- Medial heel pain with burning quality; maximal tenderness more medial/proximal than fascia origin alone.
- Chronic cases: look for ADM atrophy on MRI.
- Screen S1 and tarsal tunnel concurrently.
- If classic fasciitis care fails, reopen neuropathic heel differentials.

Imaging / confirmation

MRI may show ADM atrophy; diagnosis often clinical.

49. Complex Regional Pain Syndrome (CRPS) — Early

Disproportionate pain syndrome

What it actually is

Disproportionate regional pain with sensory, vasomotor, sudomotor, and/or motor-trophic changes after trauma or surgery.

Why it is tricky

Early CRPS is labeled “low pain threshold” or delayed healing; intervention window is missed.

Commonly mistaken for

Normal post-fracture pain, infection, arterial insufficiency, malingering (incorrect)

Correct detection steps

- Pain disproportionate to injury stage; allodynia, color/temperature change, sweating change, edema, reduced ROM.
- Use Budapest clinical criteria; early specialist pain/rehab referral.
- Exclude infection and vascular occlusion first when signs overlap.
- Early gentle reactivation—avoid immobilization spirals.

Red flags

- Rapid trophic change
- Severe allodynia blocking care

50. Double Crush Syndrome (Proximal + Distal Nerve Compromise)

Neurologic — multi-site

What it actually is

Coexistence of proximal (root/plexus/TOS) and distal (entrapment) lesions so neither alone fully explains the deficit/symptoms.

Why it is tricky

Treating only carpal tunnel or only the neck fails; each site is “almost enough” to convince the clinician.

Commonly mistaken for

Isolated CTS, isolated cervical radiculopathy, isolated cubital tunnel

Correct detection steps

- When distal release or isolated cervical care fails, re-screen the entire neural path.

- Map mixed distributions (e.g., CTS sensory + neck aggravation + positive Spurling).
 - EMG can show dual lesions; treat both contributors.
 - Ergonomic/load factors at multiple sites matter for durable recovery.
-

Disclaimer

Educational research brief for physiotherapy/MSK clinical reasoning. Not a substitute for licensed medical assessment, emergency care, or specialty guidelines. Escalate immediately for infection, dissection, PE/DVT, posterior SC dislocation, septic joint, SCFE, and acute vascular syndromes.

